Project Seminar





Master IET International Students WS2026/27

Lesson 1: Introduction

M. Eng. Andreas Hein

E-Mail: andreas.hein@hs-wismar.de

FIW: Fachbereich Elektrotechnik und Informatik

HOCHSCHULE WISMAR
UNIVERSITY OF APPLIED SCIENCES TECHNOLOGY, BUSINESS AND DESIGN

Study Regulations - University of Applied Sciences Wismar (1)

- ***Seminar implementation:***
  - The project seminar will be held in the form of consultation sessions (otherwise you will be informed in time).

- ***Registration:***
  - register for the project seminar on **Stud.IP**, where you will also find additional information and materials

https://studip.hs-wismar.de

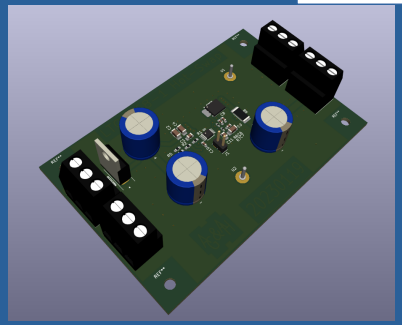
Study Regulations - University of Applied Sciences Wismar (2)

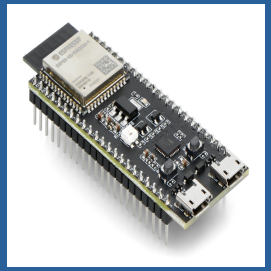
- ***Learning objectives and competences:***
  - **ability to work independently** on electrical engineering tasks with project character
  - **to work individual on projects** and practical tasks of the respective field of competence are **independently worked on**
- ***Study Exam:***
  - **Examination:** pcb project, project report (documentation) and a written test

Study Regulations (3) – Examination

PCB design

KiCad





solutions

Project documentation

Project report as PDF-file

- *Schematics, BOM, manufacturing, assembly,*
- *Solutions to tasks & exercises, etc.*

Put all parts together (PDF & all project releated folders) as Zip-File(s)!

Name, Matr.-No., face-photo of the student on the project cover on cover!

Study Regulations (4) - Exam

- ***Submission date of all project related files and documents:***

Saturday 16th January 2027 → deadline !

Project Seminar 2026





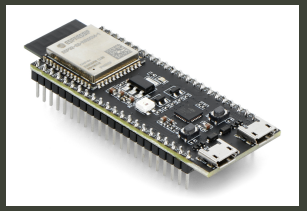
Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 1: Introduction

HOCHSCHULE WISMAR
UNIVERSITY OF APPLIED SCIENCES TECHNOLOGY, BUSINESS AND DESIGN

FIW: Fachbereich Elektrotechnik und Informatik

Project Seminar - Content



Project Seminar

Board design in KiCad

- *design pcb*
- *independently working on your project task*

Circuit analysis

- *Measurement techniques*
- *Operational Amplifier circuits*

Software development

- *become familiar with micro-controllers (µC)*
- *Dev. operational monitoring board software*
- *independently working on different programming tasks*

Project Seminar - What's needed ? (1/2)

- ***Notebook and external mouse (no touchpad !):***
  - Your notebook should use Windows 11 or Linux distributions as operating system. In case of MacOS a virtual machine from university server can be used.

  - Access to virtual machines:

IT-Info - Hochschule Wismar

Project Seminar - What's needed ? (2/2)

- ***KiCad EDA version 10.0.6 (or older stable version 9.x):***
  - You can download KiCad from:



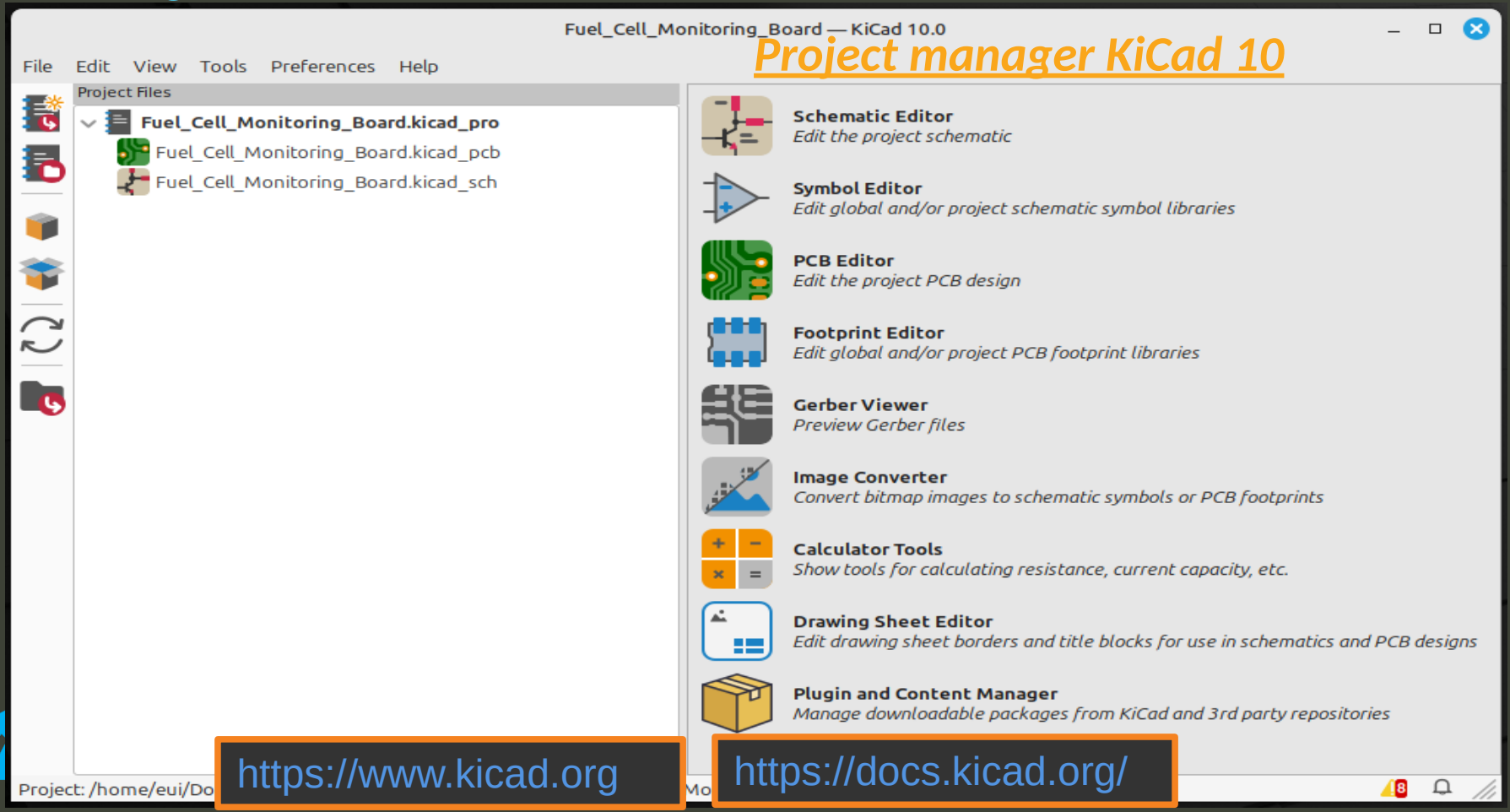
https://www.kicad.org/

Project Seminar - What is KiCad ? (1/8)

- Open source software suite for Electronic Design Automation (EDA)
- KiCad was first released in 1992 by its original author, Jean-Pierre Charras
- It has been in continual development since then and now managed by the KiCad Development Team & it is constantly being developed
- KiCad can handle Schematic Capture and PCB Layout with Gerber output.

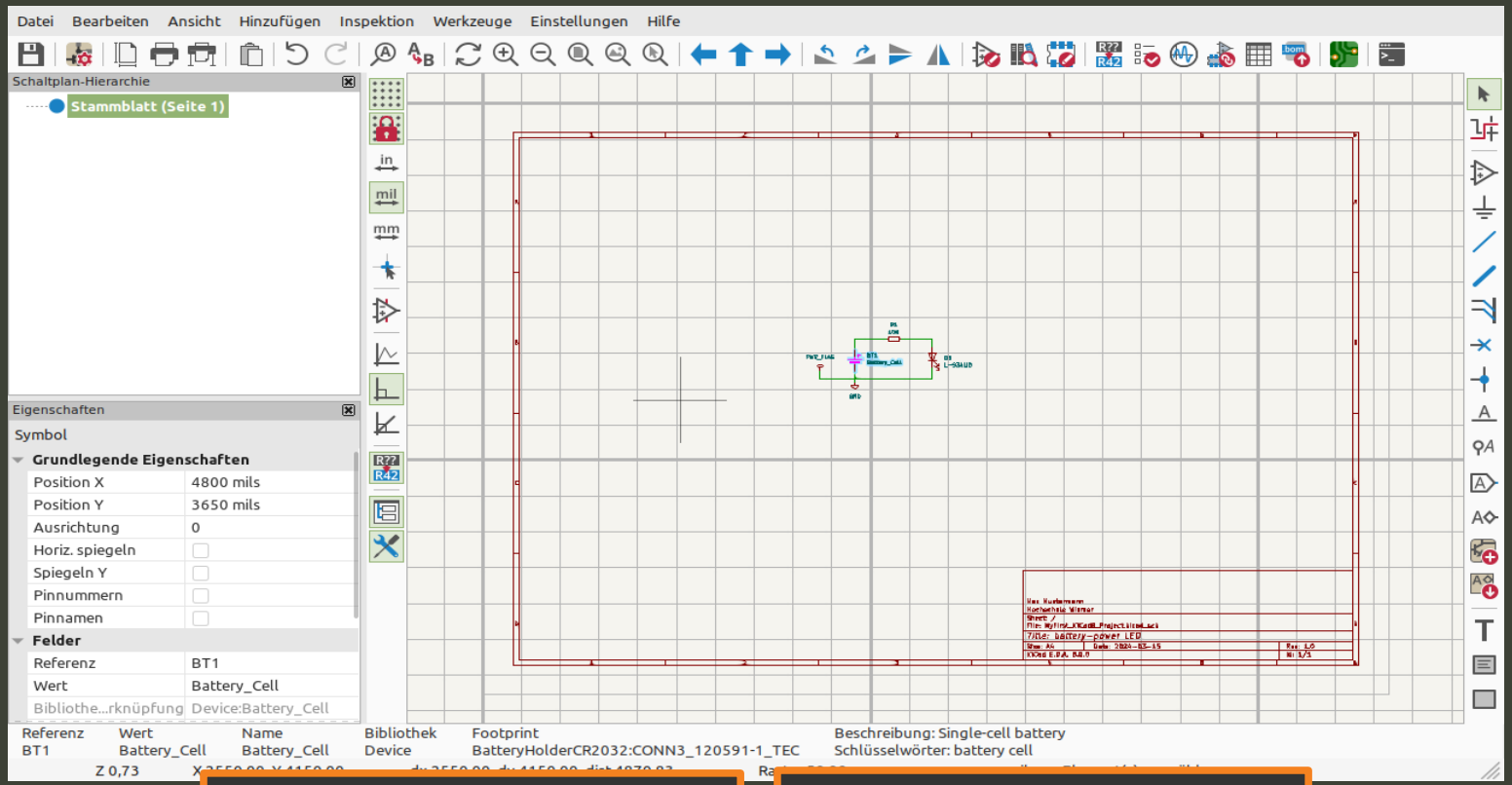


Project Seminar – What is KiCad ? (2/8)



Project Seminar – What is KiCad ? (3/8)

Schematic Editor KiCad 10

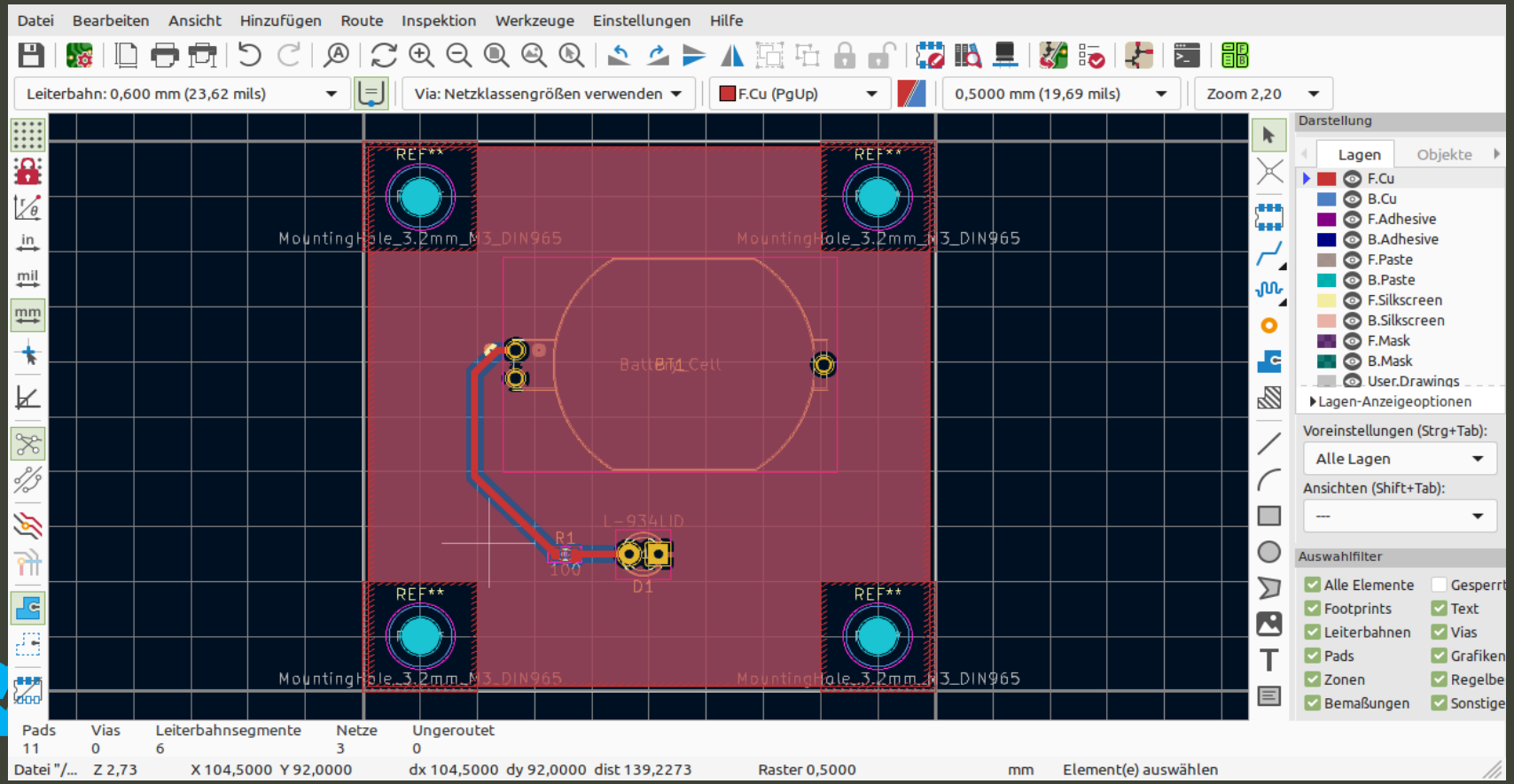


https://www.kicad.org

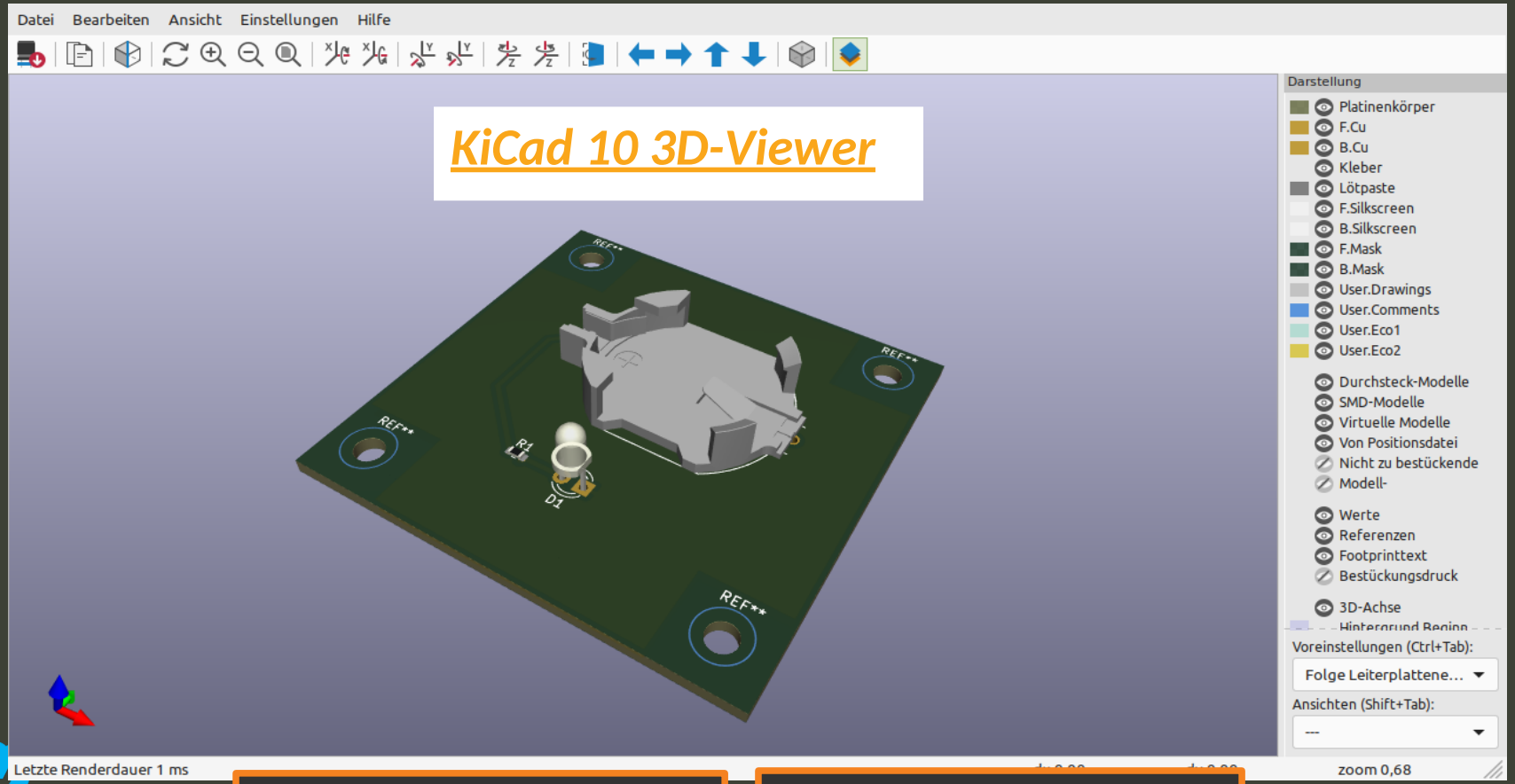
https://docs.kicad.org/

Project Seminar – What is KiCad ? (4/8)

PCB Layout Editor KiCad 10



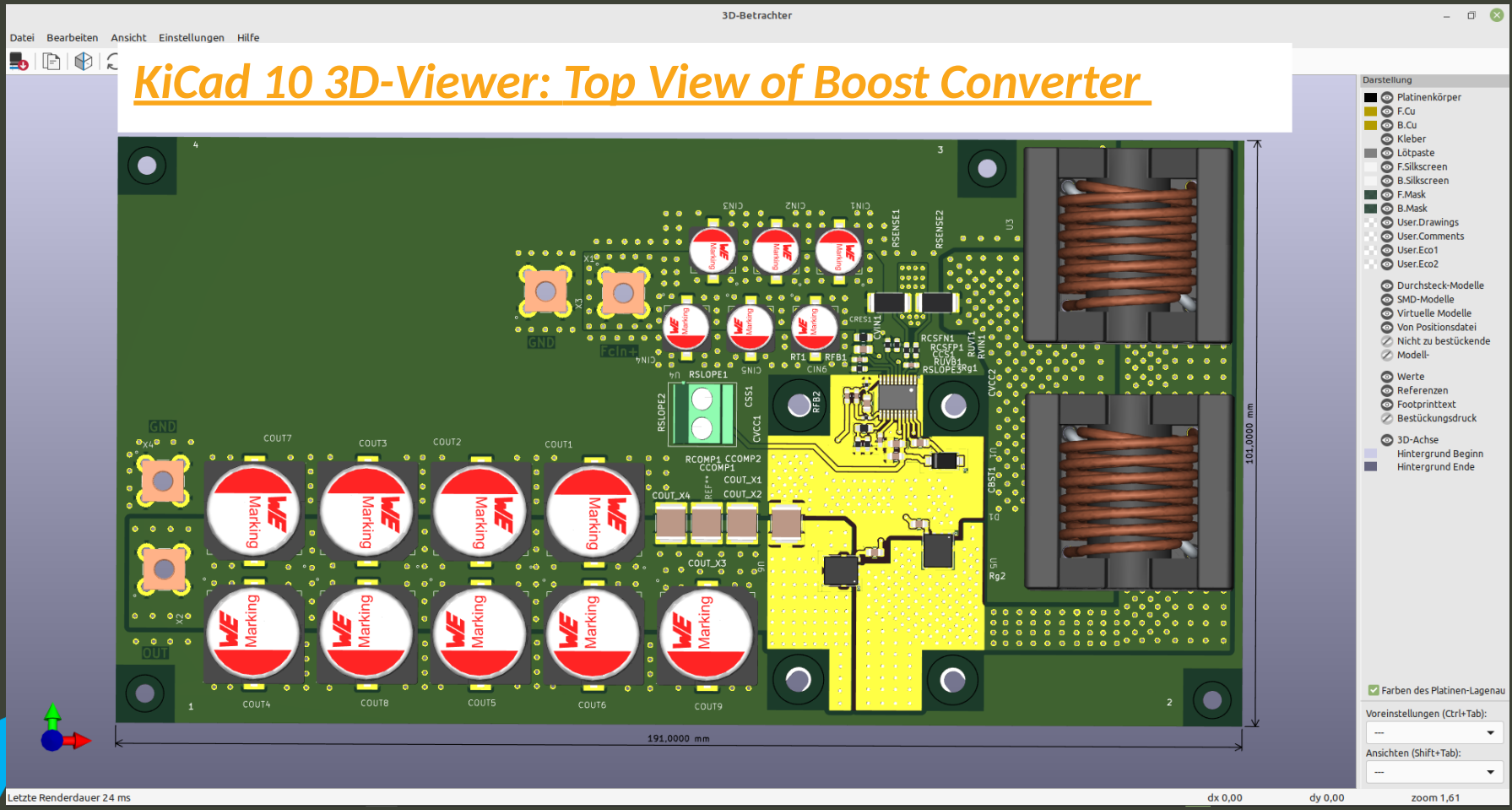
Project Seminar – What is KiCad ? (5/8)



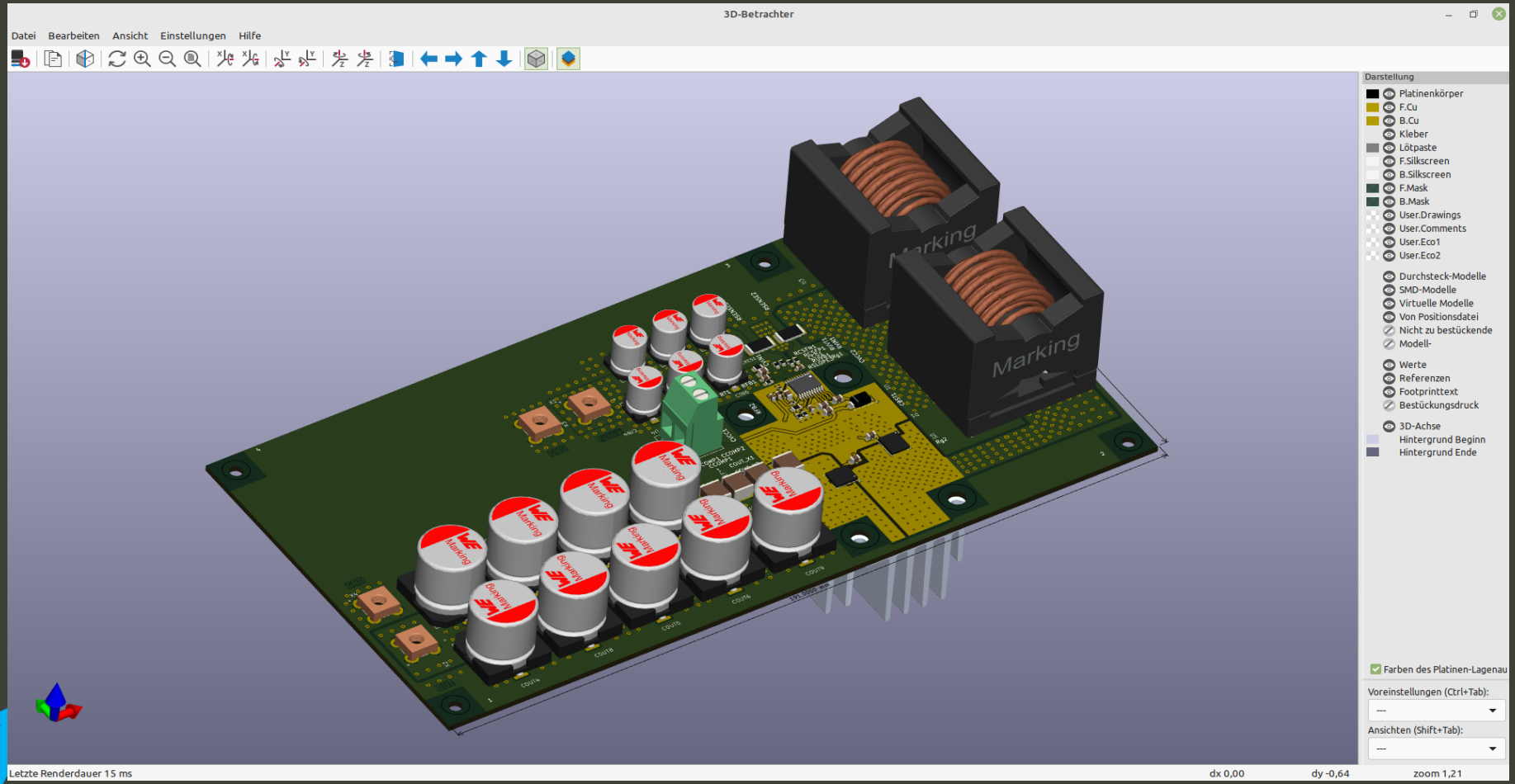
https://www.kicad.org

https://docs.kicad.org/

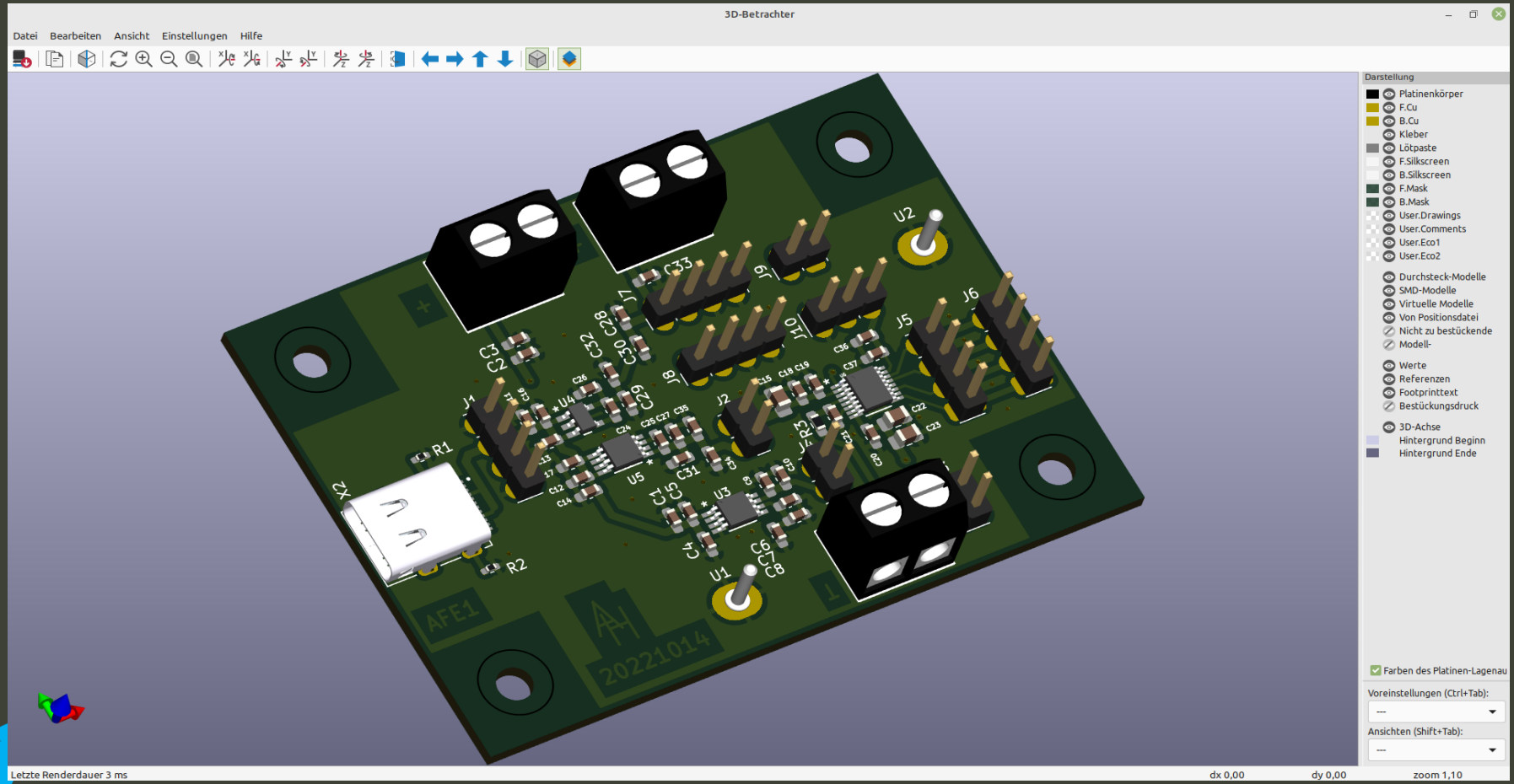
Project Seminar – What is KiCad ? (6/8)



Project Seminar – What is KiCad ? (7/8)



Project Seminar – What is KiCad ? (8/8)



Project Seminar -
What to do until next lesson ? (1/6)





- ***KiCad EDA version 10.0.6 (or older stable version 9.x.x):***
  - Install KiCad on your notebook!

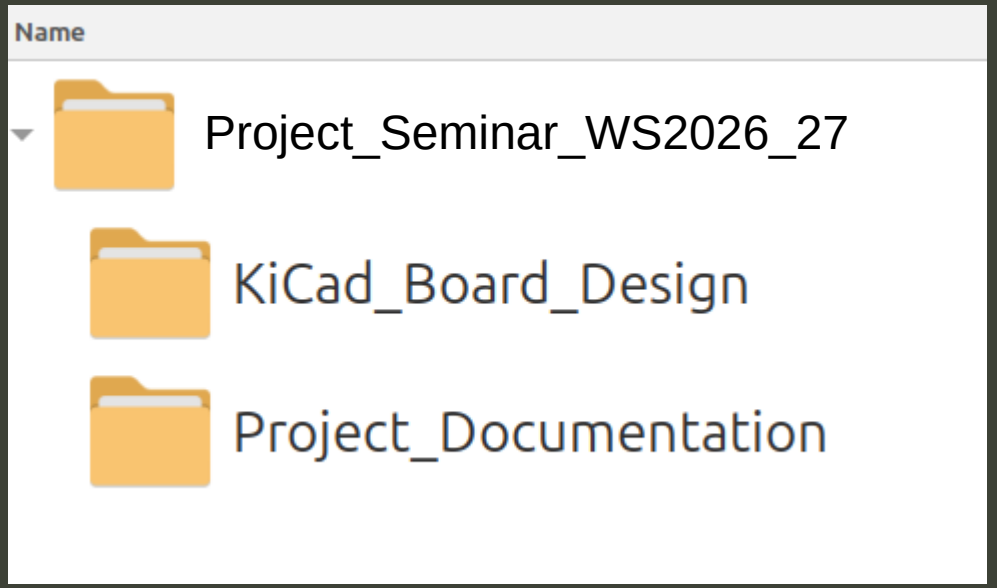
    https://www.kicad.org/
  - Make yourself familiar with KiCad – Read the documentation carefully !

    https://docs.kicad.org/
  - Create a project main-folder Project_Seminar_WS2026_27 !
  - Create in the project main-folder and two sub-folders KiCad_Board_Design and Project_Documentation !



Project Seminar -
What to do until next lesson ? (2/6)

- ***Top level folder structure:***



Project Seminar -
What to do until next lesson ? (3/6)







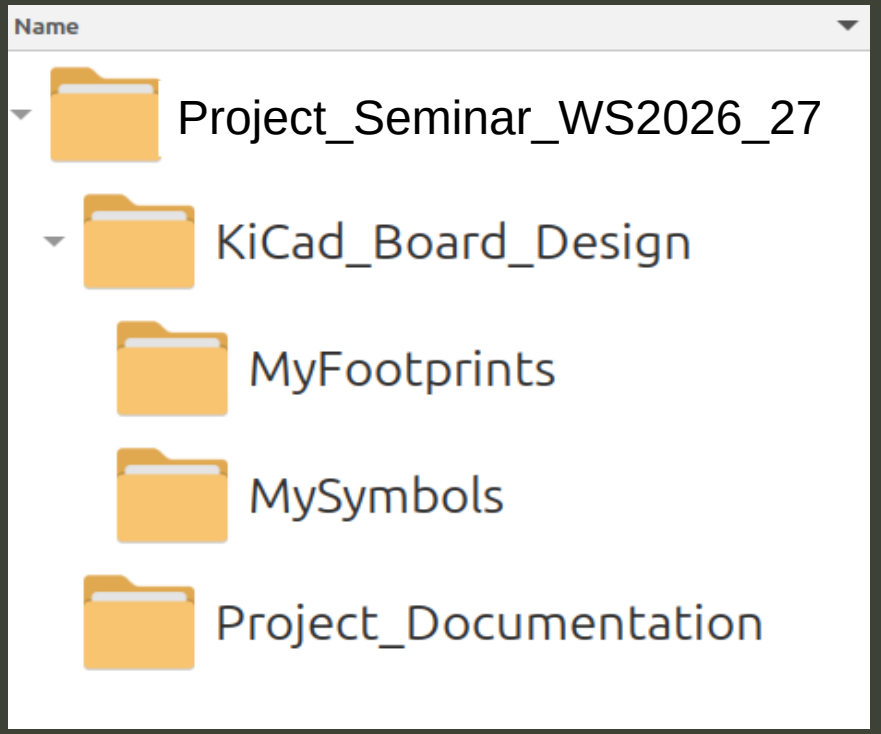
- ***Prepare a few more things:***
  - Create the following sub-folders in your KiCad_Board_Design folder with the names:
    → MySymbols
    → MyFootprints





Project Seminar -
What to do until next lesson ? (4/6)

- ***Top level folder structure → KiCad_Board_Design:***



Project Seminar -
What to do until next lesson ? (5/6)



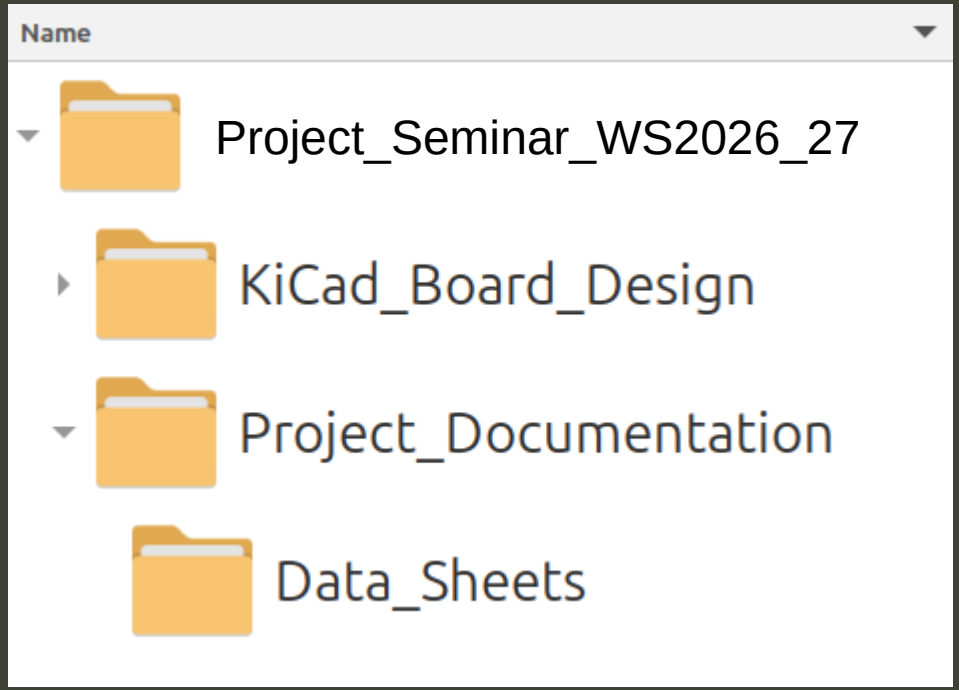


- ***Prepare a few more things:***
  - Create the following sub-folder in your Project_Documentation folder with the name:
    → Data_Sheets



Project Seminar -
What to do until next lesson ? (6/6)

- ***Top level folder structure → Project_Documentation:***



Project Seminar - Schedule (1/1)



- ***Tasks and preview:***
  - register in STUD.IP → Project Seminar
  - download and installation of KiCad 10
  - create the folder structure for project seminar
  - download KiCad 10 documentation
  - read manuals for KiCad
  - Lesson 2 – Pcb manufacturing and pcb components

… schedule is continously updated!

Project Seminar - WS2026/27



Thank you !
Have a nice day!



